

Open Letter

The general letter - the same form as the individual company letters, without claiming research into any particular company.

To the robotics, automation, AI, and physical-systems teams I meet at RoboBoston: thank you for being here, for sharing what you're building, and for taking the time to consider this.

I did not arrive at robotics by deciding to abandon the technical work I had already learned. I arrived there because, after years of moving through increasingly complex systems, those different disciplines kept revealing the same underlying structure.

Audio engineering was where that structure first became explicit to me. Working with signal chains taught me to stop seeing equipment as isolated devices and start seeing systems as paths: sources feeding transformations, control signals determining behavior, interfaces handing information from one stage to the next, and routing decisions creating consequences far downstream.

Troubleshooting became an exercise in topology. Where did the signal originate? What should happen to it next? What transformed it? What controlled that transformation? Where did the observed behavior first diverge from the expected behavior?

That way of thinking followed me everywhere else.

Mechanical and automotive work taught me to trace physical causality through engines, drivetrains, electrical faults, sensors, and failure modes. Metal fabrication and welding taught me that technical ideas eventually have to survive material reality: fit, access, sequence, tooling, force, heat, repairability, and the difference between what should work and what actually does.

Later environments made the pattern harder to ignore. At Level99, I worked around interactive systems combining RFID and other sensors, PLC-monitored elements, DMX, Arduino hardware, programmable LEDs, audio, mechanical components, and computer control. The important part was not any one technology. It was the interaction between them.

At the Starlink facility in Bastrop, Texas, I worked on clean-room structures for PCB manufacturing and saw advanced industrial systems operating at a scale where infrastructure, sequencing, integration, and reliability could not be separated. I later completed Martin Electrical School's Journeyman One program, adding formal electrical theory, code, and safety to a technical foundation that had already been developing through mechanical work, fabrication, troubleshooting, and systems thinking.

Software and AI extended the same pattern again. The medium changed, but the questions did not: What enters the system? What transforms it? Which layer controls what happens next? What context survives an interface? What dependency failed? What evidence tells you whether the problem was reasoning, routing, execution, or physical reality?

My work history and technical interests make more sense to me as one trajectory than as a collection of unrelated jobs. Audio engineering taught me to see signal paths. Mechanical diagnosis taught me to trace physical causality. Fabrication taught me to respect material reality. Interactive systems taught me to reason across sensors, controls, software, and mechanisms. Electrical study added power, protection, and formal physical-system constraints. Software and AI extended the same reasoning into information, orchestration, and decision systems. Robotics is where those layers become consequential at once.

I am not presenting myself as an experienced robotics engineer, controls engineer, embedded-firmware specialist, or someone who has already mastered the entire robotics stack. I have not.

What I can bring immediately is hands-on fabrication and welding experience, mechanical troubleshooting, electrical fundamentals, experience around sensor-driven interactive systems, comfort with tools and equipment, signal-path reasoning from audio engineering, software and AI systems thinking, and a strong habit of tracing causes across domain boundaries.

I am interested in roles where that foundation is useful now and where I can continue developing deeper professional competence in robotics hardware, electromechanical integration, test, reliability, controls, prototype build, field support, and systems that have to work outside of a diagram.

If that combination is useful to your team, I would like to talk.

Zachary Neagle